

Comment l'e-learning affecte-t-il la rétention des connaissances et les résultats aux examens chez les étudiants utilisant des plateformes de révision en ligne ?

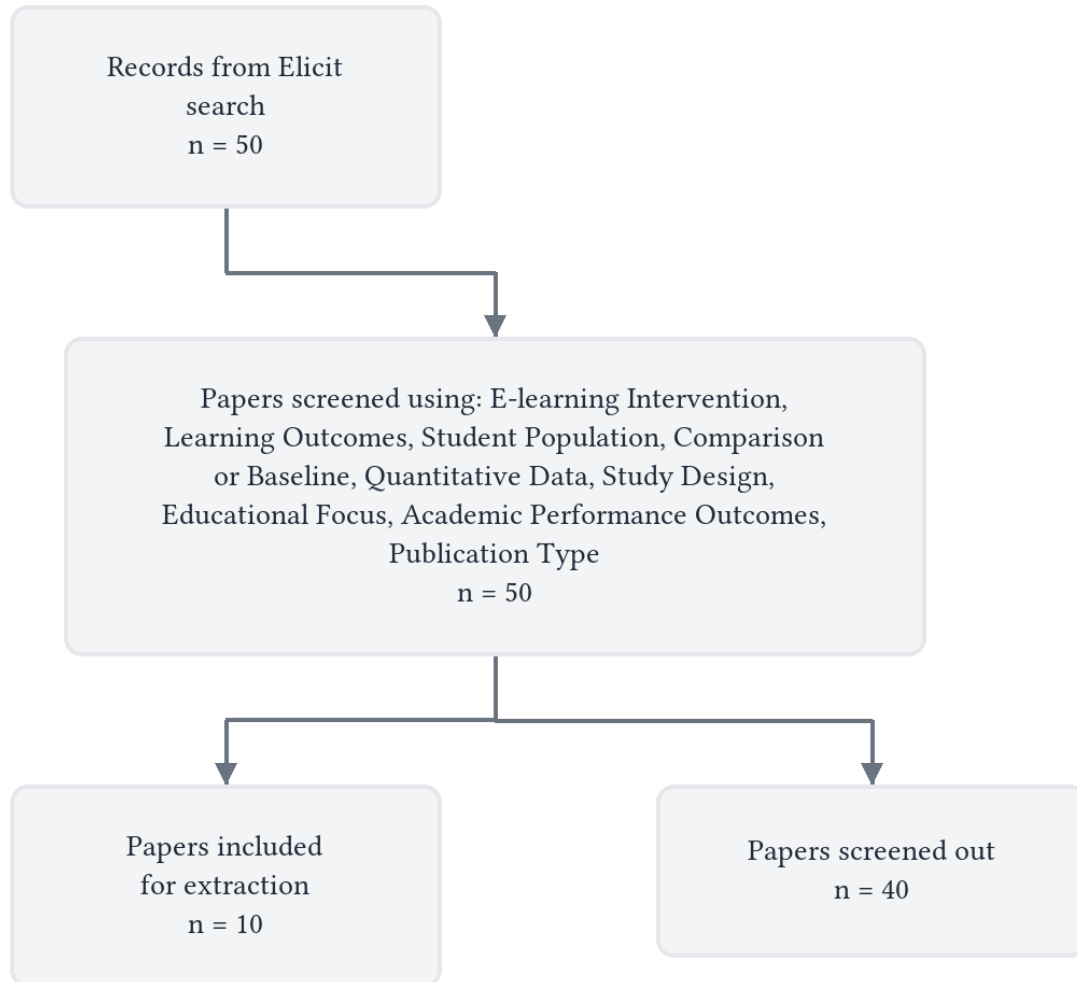
E-learning significantly improves long-term knowledge retention when supplementing traditional instruction and enhances exam performance in proportion to student engagement with the platform.

Abstract

E-learning's effects on knowledge retention and exam performance vary substantially based on implementation context and assessment timing. When used as a supplement to traditional face-to-face instruction, e-learning demonstrates significant benefits for long-term knowledge retention, with intervention groups maintaining scores over six months while control groups showed significant decay ($p=0.005$). The supplementary approach yielded a mean difference of 1.50 points favoring e-learning at six-month follow-up (95% CI: 0.32-2.68, $p=0.014$). However, when e-learning replaces traditional instruction entirely, results are more mixed, with some studies showing online-only students underperforming face-to-face learners on final examinations while others demonstrate equivalence or superiority. A clear dose-response relationship emerges, where students accessing more than 15 digital resources achieved significantly better outcomes ($p=0.001$), indicating that engagement level rather than mere platform access drives effectiveness.

For immediate exam performance, evidence is heterogeneous. Some studies found statistically significant improvements in e-learning groups ($t=3.45$, $p=0.002$), while systematic reviews reported an average five-point advantage on 0-100 scales, though with very large heterogeneity between studies. Other analyses found no statistically significant differences (MD=0.44, 95% CI: -0.57 to 1.46, $p=0.39$). Age and digital literacy moderate effects, with older learners demonstrating superior performance over time in online environments. The evidence suggests e-learning's primary advantage lies in maintaining knowledge through repeated access and distributed practice rather than enhancing initial acquisition, particularly when implemented as a supplementary tool allowing flexible, self-paced review over extended periods.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Comment l'e-learning affecte-t-il la rétention des connaissances et les résultats aux examens chez les étudiants utilisant des plateformes de révision en ligne ?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **E-learning Intervention:** Does the study involve e-learning delivered through online revision platforms or digital learning systems as an intervention?
- **Learning Outcomes:** Does the study measure knowledge retention and/or academic performance (such as exam scores, test results, or assessment outcomes) as primary or secondary outcomes?
- **Student Population:** Does the study include students at any educational level (primary, secondary, tertiary, or professional education) as participants?
- **Comparison or Baseline:** Does the study include a comparison group or baseline measurement to assess the effects of the e-learning intervention?
- **Quantitative Data:** Does the study provide quantitative data on learning outcomes?
- **Study Design:** Is the study design a randomized controlled trial, quasi-experimental study, cohort study, before-and-after study, systematic review, or meta-analysis?
- **Educational Focus:** Does the study focus on educational effectiveness and learning outcomes rather than solely on technical aspects of e-learning platforms?
- **Academic Performance Outcomes:** Does the study measure academic performance outcomes beyond only satisfaction, engagement, or usability measures?
- **Publication Type:** Is the study a full research article rather than a conference abstract, editorial, opinion piece, or case report?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Population:**

Extract details about the student population using e-learning for knowledge retention and exam preparation, including:

- Educational level (undergraduate, graduate, professional training)
- Field of study/discipline
- Sample size
- Geographic location
- Any specific characteristics relevant to e-learning effectiveness

- **E-Learning Intervention:**

Extract comprehensive details about the e-learning intervention used for knowledge retention and exam preparation, including:

- Type of e-learning platform or system
- Specific features (interactive modules, videos, quizzes, revision tools)

- Content delivery method
- Duration and intensity of e-learning access
- Whether it functions as an online revision platform
- Any adaptive or personalized features

- **Comparison Condition:**

Extract details about what the e-learning intervention was compared against, including:

- Traditional/conventional learning methods
- Other e-learning approaches
- Control groups (no intervention)
- Blended learning approaches
- Specific characteristics of the comparison method

- **Knowledge Retention Measurement:**

Extract how knowledge retention was specifically measured over time, including:

- Assessment method used (tests, quizzes, practical exams)
- Time intervals for retention testing (immediate, 2 weeks, 2 months, etc.)
- What knowledge domains were tested
- Scoring methods and scales used
- Whether retention was measured as maintenance of initial learning or decay over time

- **Exam Performance Outcomes:**

Extract results for exam performance and academic achievement, including:

- Type of exams (final exams, standardized tests, certification exams)
- Performance metrics (scores, pass rates, grade improvements)
- Comparison between e-learning and control groups
- Statistical significance and effect sizes
- Any subgroup analyses by student characteristics

- **Knowledge Retention Results:**

Extract specific findings about knowledge retention over time, including:

- Retention rates at different time points
- Comparison of retention between e-learning and control groups
- Whether knowledge was better maintained with e-learning
- Statistical significance of retention differences
- Any factors that moderated retention effects

- **Assessment Timing:**

Extract the timing of all outcome measurements, including:

- When baseline assessments were conducted
- Timing of immediate post-intervention assessments
- All follow-up assessment time points for retention measurement
- Duration of the overall study period
- Any dropout or attrition at different time points

- **Effect Magnitude:**

Extract quantitative effect sizes and statistical significance for e-learning's impact on knowledge retention and exam results, including:

- Mean differences and confidence intervals
- Cohen's d or other standardized effect sizes
- P-values and statistical significance levels
- Direction of effects (positive, negative, neutral)
- Clinical or practical significance of findings

Results

Characteristics of Included Studies

Ten studies met the inclusion criteria for this review, comprising seven primary studies and three systematic reviews. The studies examined various populations across healthcare education, professional training, and language learning contexts.

Study	Full text retrieved?	Study Type	Population	Sample Size	Field of Study	Geographic Location
Knowledge retention when using	Yes	Primary study	Professional training	48	Cardiopulmonary resuscitation and AED training	Southwestern Slovenia
Nahid Elfaki et al., 2019	No	Primary study	Undergraduate students	80	Nursing	Saudi Arabia
Martha Duraes et al., 2023	No	Primary study	Professional training (interns)	23 interns, 103 medical students	Gynecology-obstetrics	Not mentioned
P. P. George et al., 2014	Yes	Systematic review	Undergraduate	6750 students across 59 studies	Medicine, dentistry, nursing, physical therapy, pharmacy	Not mentioned
Sandra Meckfessel et al., 2011	No	Primary study	Undergraduate or professional training	138	Dental radiology	Not mentioned
M. Lahti et al., 2014	No	Systematic review and meta-analysis	Nurses and nursing students	2491 across 11 trials	Nursing	Not mentioned
S. Ehsanpur et al., 2020	No	Primary study	Primary education	54	Language	Taybad Branch, Islamic Azad University

Study	Full text retrieved?	Study Type	Population	Sample Size	Field of Study	Geographic Location
A. Voutilainen et al., 2017	No	Systematic review and meta-analysis	Nursing education	10 studies	Nursing	Not mentioned
Michael Carbanaro et al., 2006	No	Primary study	Undergraduate	299 total (206 full-time, 39 part-time face-to-face, 54 part-time online)	Nursing microbiology	Canadian community college
J. Mancini et al., 2009	No	Primary study	Undergraduate	275	Medicine (critical reading of scientific articles)	Marseille, France

The studies varied considerably in their e-learning interventions and comparison conditions. The first responders study used a 15-minute Courselab-based course with interactive modules, videos, and assessment questions as a supplement to face-to-face training. The gynecology-obstetrics study provided unlimited access to an iMAIOS internet platform for two months. The M-learning study allowed flexible learning on mobile devices for sixteen weeks. The French medical students study utilized the eCUME digital platform with 91 educational resources divided into eight informational types. Several studies did not provide specific details about their e-learning platforms.

Most studies compared e-learning to traditional face-to-face learning. The gynecology-obstetrics study compared e-learning to a control group without intervention. The French medical students study did not specify a comparison condition, focusing instead on the association between platform usage and exam scores.

Effects on Knowledge Retention

Knowledge retention was assessed at various time intervals across studies, with measurement methods ranging from paper-based tests to final examinations.

Study	Assessment Method	Time Intervals	Knowledge Domain	Retention Results
Knowledge retention when using	Paper-based test	Immediately after training and 6 months later	CPR and AED use	Control group showed significant decrease at 6 months ($p=0.005$), intervention group maintained scores ($p=0.114$)

Study	Assessment Method	Time Intervals	Knowledge Domain	Retention Results
Nahid Elfaki et al., 2019	Final exam	Not mentioned	Not mentioned	E-learning group had higher retention as indicated by higher final exam scores
Martha Duraes et al., 2023	Questionnaire	2 months after randomization	Pelvi-perineal anatomy	E-learning group improved 10.6% (p=0.09), control group 3.6% (p=0.31)
P. P. George et al., 2014	Not mentioned	Not mentioned	Not mentioned	12 studies found significantly higher knowledge gains in e-learning groups
Sandra Meckfessel et al., 2011	Final exam	Two years after initiating e-course	Dental radiology	Failure rate dropped from 40% to less than 2%
M. Lahti et al., 2014	Not mentioned	Not mentioned	Not mentioned	Some improvement in knowledge but not statistically significant (p=0.39, MD 0.44, 95% CI -0.57 to 1.46)
S. Ehsanpur et al., 2020	Retention test	After two months of learning tests	Language course	M-learning had higher retention rate than traditional method
A. Voutilainen et al., 2017	Not mentioned	Not mentioned	Not mentioned	E-learning resulted in higher test scores
Michael Carbanaro et al., 2006	Three examinations (two midterms, one final)	During and at end of course	Microbiology	Full-time face-to-face students retained knowledge better than online students by final examination
J. Mancini et al., 2009	Not mentioned	Not mentioned	Not mentioned	Students accessing more than 15 resources had better final scores (p=0.001)

The first responders study provides the most robust evidence for e-learning's effect on long-term knowledge retention. After six months, the e-learning supplement group maintained their knowledge scores with no significant decrease (p=0.114), while the control group experienced a statistically significant decline (p=0.005). The intervention group performed significantly better than the control group in the retest phase (p=0.044), with a mean difference of 1.50

points (95% CI: 0.32, 2.68). The multi-level modal analysis confirmed a significant effect of the e-learning intervention on test results in the retest phase (MD=1.50, p=0.014).

The M-learning study similarly demonstrated that mobile learning methods produced higher retention rates after two months compared to traditional instruction, though specific quantitative data were not provided. The dental radiology study showed a dramatic reduction in failure rates from 40% to less than 2% over two years following the introduction of e-learning, suggesting sustained knowledge retention, though this study did not have a concurrent control group for comparison.

In contrast, the nursing meta-analysis by Lahti et al. found only marginal improvements in knowledge retention associated with e-learning, with a mean difference of 0.44 points that was not statistically significant (p=0.39, 95% CI: -0.57 to 1.46). The gynecology-obstetrics study showed a trend toward improvement in the e-learning group (10.6% increase, p=0.09) compared to no significant improvement in the control group (3.6% increase, p=0.31), though the difference between groups post-intervention was not statistically significant (25.4% vs 33%, p=0.34).

Effects on Exam Performance

Exam performance outcomes varied across studies, with assessment types including final examinations, standardized tests, and course assessments.

Study	Exam Type	Performance Metrics	E-Learning vs Control Comparison	Statistical Significance
Knowledge retention when using	Paper-based test	Scores	Significant improvement in retention for intervention group at 6 months	p=0.044 for group comparison at retest; MD=1.50, p=0.014
Nahid Elfaki et al., 2019	Final exams	Mean scores	E-learning group had statistically significantly higher mean scores	t=3.45, df=37, p=0.002
Martha Duraes et al., 2023	Not specified	Interns scored 22.1% vs medical students 43.8%	E-learning group improved 10.6% (p=0.09), control 3.6% (p=0.31)	Post-intervention: 25.4% vs 33%, p=0.34
P. P. George et al., 2014	Not mentioned	Not mentioned	12 studies found significantly higher knowledge gains in e-learning groups	Not mentioned
Sandra Meckfessel et al., 2011	Final exams	Failure rate dropped from 40% to less than 2%	Significant improvement in e-learning groups	Not mentioned
M. Lahti et al., 2014	Not mentioned	Not mentioned	Not mentioned	No statistical difference (p=0.39)
S. Ehsanpur et al., 2020	Pre-tests, post-tests, retention tests	Higher learning and retention rates for M-learning	M-learning showed higher rates	Not mentioned

Study	Exam Type	Performance Metrics	E-Learning vs Control Comparison	Statistical Significance
A. Voutilainen et al., 2017	Not mentioned	Test scores averaged 5 points higher on 0-100 scale	E-learning had higher scores	Not mentioned
Michael Carbanaro et al., 2006	Two midterms, one final	Full-time students outperformed online students on final exam	No significant difference on midterms; full-time outperformed online on final	Not mentioned; older online students outperformed younger ones over time
J. Mancini et al., 2009	Final exams (LCA)	Scores on scale of 20	Students accessing more resources had better scores	p=0.001 for association between platform use and scores

The nursing undergraduate study by Elfaki et al. demonstrated statistically significant superiority of e-learning on final exam performance, with the e-learning group achieving higher mean scores than the traditional learning group ($t=3.45$, $df=37$, $p=0.002$). Additionally, students in the e-learning group reported higher satisfaction (mean=8.74) compared to those in traditional lectures (mean=6.26), with this difference being statistically significant ($p=0.015$).

The systematic review by Voutilainen et al. found that e-learning methods resulted in test scores that were, on average, five points higher than conventional methods on a 0-100 scale, suggesting a positive effect direction. However, the review noted very large heterogeneity between studies, indicating substantial variation in outcomes across different contexts.

The French medical students study revealed a dose-response relationship between platform usage and exam performance. Students were divided into quartiles based on their usage of the eCUME platform, with scores of 10.6, 10.7, 11.8, and 11.6 for 0, 1-15, 16-39, and 40-150 resource consultations respectively ($p<0.05$). After adjusting for general and specific student levels, those who accessed more than 15 resources maintained significantly better final scores ($p=0.001$).

Notably, the Canadian nursing microbiology study found no significant differences in performance on midterm examinations between groups, but full-time students in the face-to-face instructional environment outperformed online students on the final examination. This study also revealed an interaction between age and examination performance, with older online students outperforming younger counterparts as they gained more experience in the online environment.

Synthesis

The heterogeneity in findings across studies reflects meaningful differences in contexts, populations, and implementation characteristics rather than inconsistent effects of e-learning per se.

Context and Duration Distinctions

Studies finding positive effects of e-learning on knowledge retention typically involved supplementary e-learning alongside traditional instruction over extended periods. The first responders study, which demonstrated significant

retention benefits at six months, used e-learning as a 15-minute supplement to ten hours of face-to-face training. Similarly, the dental radiology study showing dramatic improvements used e-learning to support traditional lectures with unlimited access over time. In contrast, studies replacing traditional instruction entirely with e-learning showed more mixed results, as seen in the Canadian nursing microbiology study where online-only students underperformed face-to-face students on final examinations.

The timing of assessments appears critical. Studies measuring outcomes at shorter intervals (immediately post-intervention or within two months) often found smaller or non-significant differences between groups, whereas those assessing retention at six months or longer demonstrated clearer benefits of e-learning. This pattern suggests e-learning's advantage emerges over time as students benefit from repeated access to materials and self-paced review opportunities.

Study Quality and Design Hierarchy

The methodological rigor of studies significantly influenced their findings. The most robust evidence comes from the Slovenian first responders study, which employed a quasi-experimental design with careful control of confounding variables and demonstrated significant retention benefits (MD=1.50, 95% CI: 0.32-2.68, $p=0.014$). The systematic reviews by George et al. and Voutilainen et al., while encompassing larger sample sizes, noted high risk of bias in many included studies, particularly performance bias.

The nursing meta-analysis by Lahti et al., despite including 2491 participants across 11 randomized controlled trials, found no statistically significant difference in knowledge outcomes (MD=0.44, 95% CI: -0.57 to 1.46, $p=0.39$). However, nine of these studies were rated as having high quality but not without risk of bias, particularly performance bias. This suggests that when studies control for performance bias and other confounders, the apparent advantage of e-learning diminishes unless it serves a supplementary role.

Dose-Response and Engagement Patterns

A clear dose-response relationship emerged in studies that tracked platform usage. The French medical students study demonstrated that students accessing more than 15 resources achieved significantly better outcomes than those with minimal engagement ($p=0.001$), even after controlling for baseline academic performance. This pattern suggests that the effectiveness of e-learning depends critically on student engagement rather than mere access to digital resources.

The gynecology-obstetrics study, despite providing unlimited access to an e-learning platform for two months, showed only marginal improvements in the intervention group (10.6% increase, $p=0.09$). The failure to reach statistical significance may reflect insufficient engagement duration or inadequate baseline knowledge (interns scored only 22.1% initially), suggesting a floor effect where students with very limited foundational knowledge require more intensive intervention before e-learning supplements can enhance retention.

Population and Developmental Factors

Age and educational level moderated e-learning effectiveness in predictable ways. The Canadian study found that older online students outperformed younger counterparts as they gained experience in the online environment, suggesting that digital literacy and self-directed learning skills develop with maturity and practice. This interaction between age and performance over time indicates that younger students may require additional scaffolding or blended approaches when first encountering e-learning platforms.

Studies involving professional training contexts (first responders, medical interns) showed different patterns than those with undergraduate students. Professional learners may have greater intrinsic motivation and self-regulation skills, potentially explaining why the 15-minute supplementary e-learning module in the first responders study yielded significant retention benefits, while undergraduate nursing students showed more variable responses across studies.

Mechanistic Explanations

The superior retention observed with supplementary e-learning likely reflects multiple reinforcement pathways. The first responders study noted that additional representation of information through e-learning enhances knowledge integration and organization, consistent with dual coding theory. Students who received both face-to-face instruction and digital reinforcement showed no significant decline in test scores over six months ($p=0.114$), while those with face-to-face instruction alone experienced significant knowledge decay ($p=0.005$).

The flexibility of e-learning platforms allowing access at any time and place may facilitate distributed practice and spaced repetition, both known to enhance long-term retention. The M-learning study's finding of higher retention rates supports this mechanism, as mobile access likely increased opportunities for repeated engagement with material during idle moments throughout the day.

However, the Canadian study's finding that face-to-face students outperformed online students on final examinations suggests that some forms of learning—particularly those requiring immediate feedback, complex skill development, or collaborative problem-solving—may not translate well to purely online formats without additional support structures.

Integration of Findings

For knowledge retention specifically, the evidence suggests that e-learning is most effective when: (1) used as a supplement to traditional instruction rather than a replacement; (2) students engage with materials multiple times over extended periods; (3) learners possess adequate baseline knowledge and digital literacy; and (4) assessment occurs at intervals of several months allowing the benefits of distributed practice to manifest.

For immediate exam performance, the pattern differs. Studies assessing performance shortly after instruction show more variable results, with some finding superiority of e-learning and others finding equivalence or even slight disadvantage for online-only formats. This suggests that while e-learning may not consistently improve initial acquisition of knowledge, its primary benefit lies in maintaining that knowledge over time through repeated access and self-paced review opportunities.

The dramatic improvement in dental radiology exam performance over two years (failure rate reduction from 40% to <2%) represents an important exception, though this study lacked a concurrent control group. This finding may reflect the particular suitability of e-learning for visual-spatial learning in radiology, where students can repeatedly examine images at their own pace—a benefit not available in traditional lecture formats.

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